Smoothing ladder of the best evolved design

`re_w406t3_r01_p10_c16`: $W_{\mathrm{ULTRALIGHT_v1}} = 80.299390$ as exported, ten smoothing levels

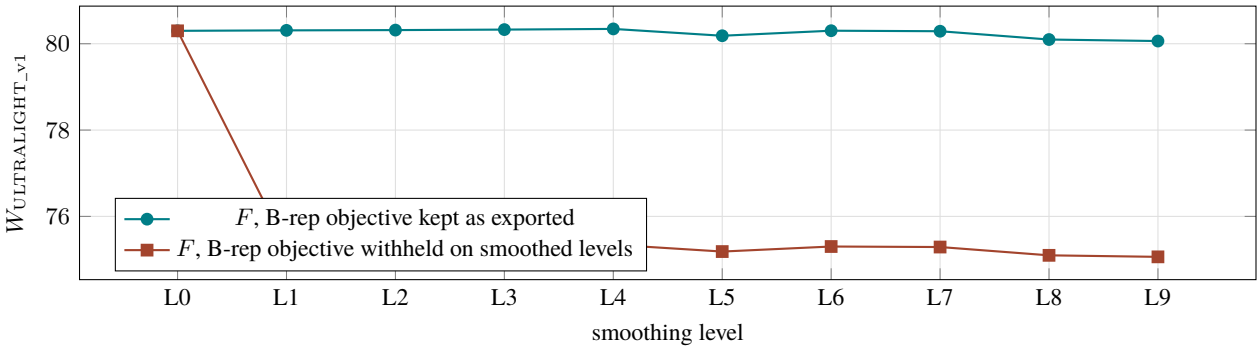
Research draft; author approval pending. Generated by `make-Cdocs/papersmoothing-ladder`; receipt `docs/paper/ladder/ladder.json`.

What this shows. The exported triangle mesh of the design is smoothed at ten strengths, from none (L0) to the strongest setting the smoother completes on its knife edges without reversing a face (L9). Each level is re-scored with the canonical `W_ULTRALIGHT_v1` and rendered from one fixed oblique camera under one low raking light, with the thickness direction exaggerated and faces coloured by height, so that grinds, serrations, channels and faceting stand out. Every camera, light, colour range and exaggeration is shared by all levels; image differences are mesh differences.

Smoothing. Taubin λ/μ filtering ($\lambda = 0.5$, $\mu = -0.5$), keeping only the component of each umbrella vector along the area-weighted vertex normal, total displacement capped at 2 mm. Vertices on edges whose two face normals differ by more than the pin angle do not move. L1–L5 pin at 30° (L4, 20 iterations, is the R1 setting of paper variant B); L6–L9 pin at 60°, the weakest pin angle that completes on this mesh.

Re-scoring. Three of the twenty objectives read the exported 3D geometry and are re-measured at every level: 3D mass (enclosed volume times $4.43\,\mathrm{g\,cm^{-3}}$), the solid tip (scikit-fem P2 submodel on the smoothed skin, cross-checked by DOLFINx 0.9.0 on the same mesh) and the CAD/B-rep objective. The B-rep objective is withheld (scored 0) on every smoothed level, because a smoothed triangle mesh has no validated STEP solid; that costs 5 points at once, so the table also gives F with the B-rep objective kept as exported, which isolates the effect of smoothing. The other seventeen objectives are computed from the genome and carry over unchanged: in particular, edge sharpness and point profile are *not* re-measured on the smoothed mesh. Neither score is a physical test.

Level	Pins	Iter.	Max disp. (mm)	RMS disp. (mm)	Vol. (%)	Mass (g)	Tip SF (DOLFINx)	F	F (B-rep kept)
L0	–	0	–	–	–	83.2231	0.6823	80.299390	80.299390
L1	30°	2	0.101	0.0056	-0.014	83.2113	0.6848	75.309944	80.309944
L2	30°	5	0.166	0.0093	-0.032	83.1962	0.6856	75.316252	80.316252
L3	30°	10	0.194	0.0123	-0.057	83.1758	0.6877	75.327832	80.327832
L4	30°	20	0.214	0.0156	-0.091	83.1472	0.6905	75.343635	80.343635
L5	30°	60	0.252	0.0222	-0.153	83.0962	0.6333	75.185235	80.185235
L6	60°	20	0.423	0.0361	0.015	83.2352	0.6842	75.301958	80.301958
L7	60°	60	0.495	0.0502	-0.058	83.1745	0.6750	75.290069	80.290069
L8	60°	120	0.613	0.0636	-0.214	83.0448	0.5997	75.097738	80.097738
L9	60°	200	0.703	0.0766	-0.399	82.8909	0.5747	75.062571	80.062571



Colour scales. Height above the mid-plane, 0 to the stock half-thickness 2.61 mm (thicker parts such as the finger ring saturate):



0 mm 2.61 mm

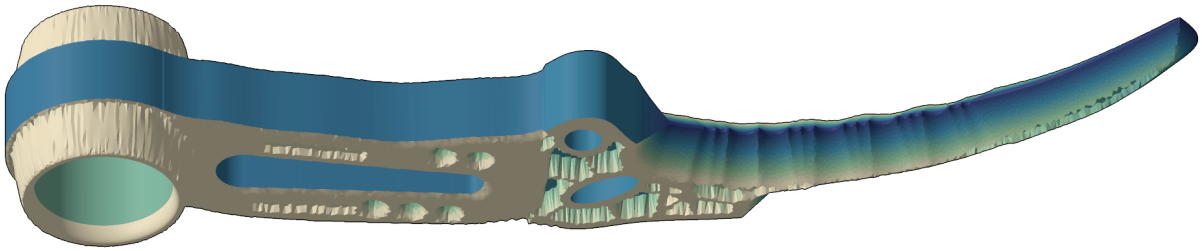
Displacement from L0, shared by all levels, 0 to 0.633 mm:



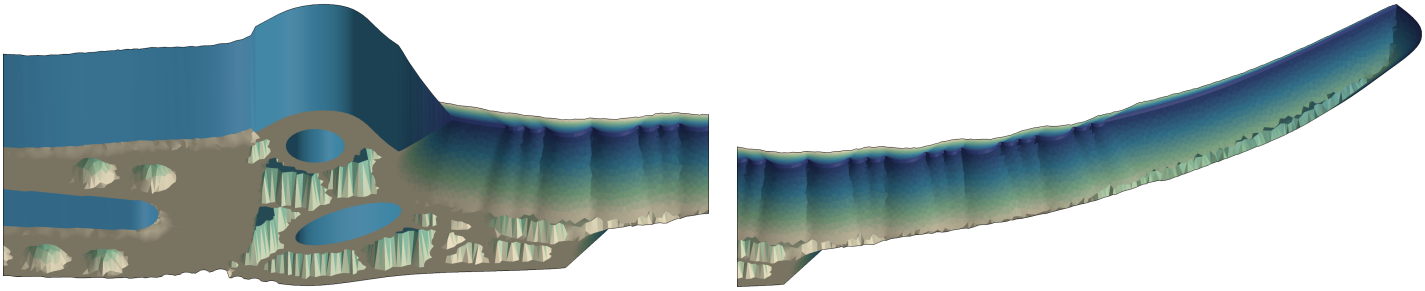
0 mm 0.633 mm

Level L0 as exported (no smoothing)

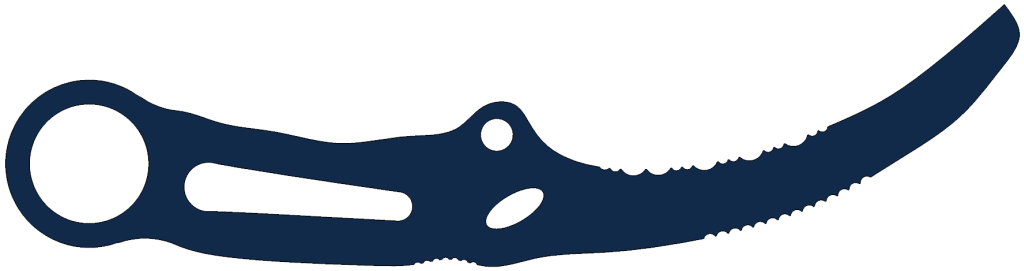
$F = 80.299390$ (with B-rep kept: 80.299390) max displacement 0.000 mm, RMS 0.0000 mm tip SF 0.6823 3D mass 83.2231 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



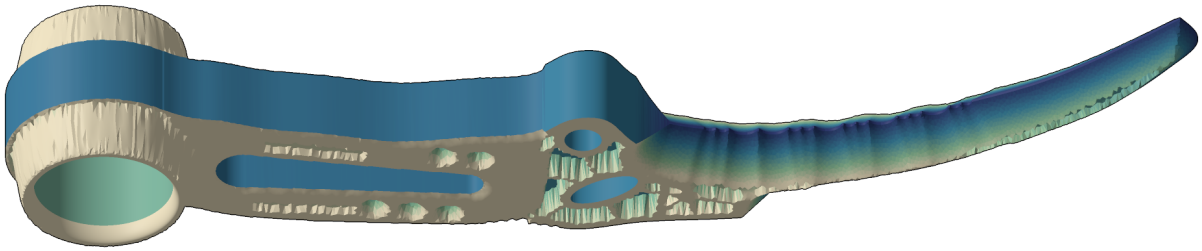
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



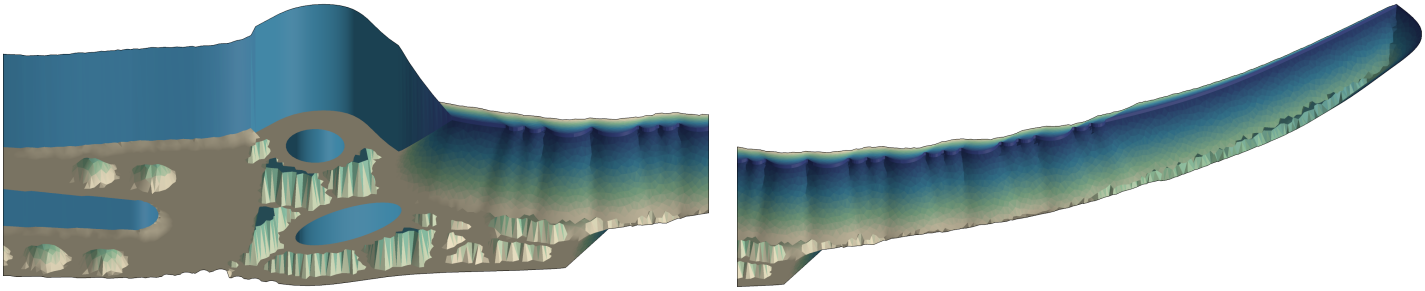
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L1 2 iterations, feature pins 30°

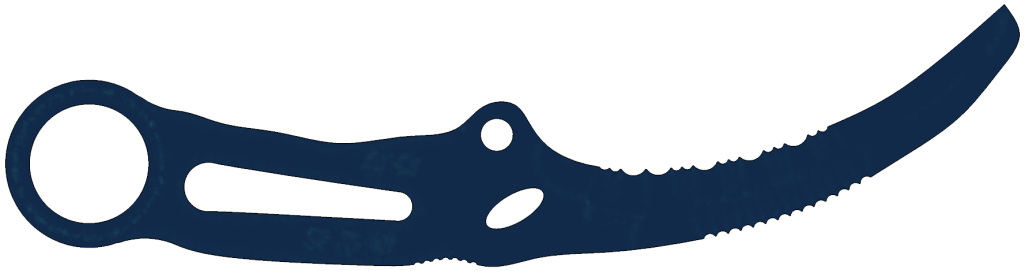
$F = 75.309944$ (with B-rep kept: 80.309944) max displacement 0.101 mm, RMS 0.0056 mm tip SF 0.6848 3D mass 83.2113 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



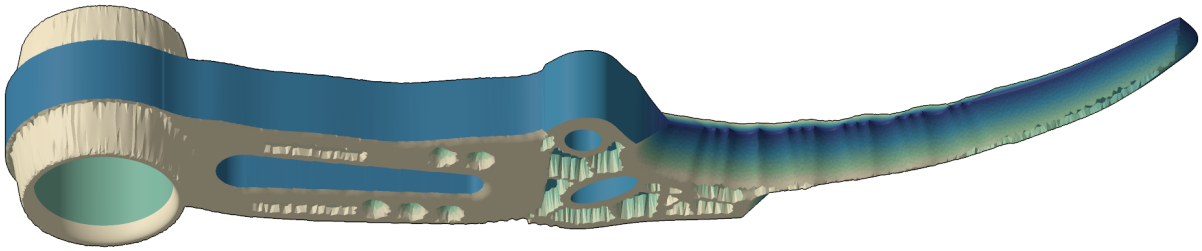
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



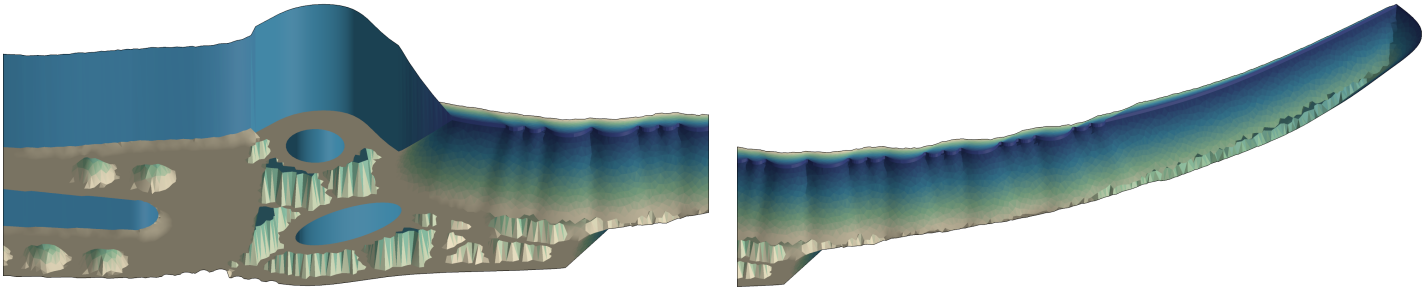
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L2 5 iterations, feature pins 30°

$F = 75.316252$ (with B-rep kept: 80.316252) max displacement 0.166 mm, RMS 0.0093 mm tip SF 0.6856 3D mass 83.1962 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



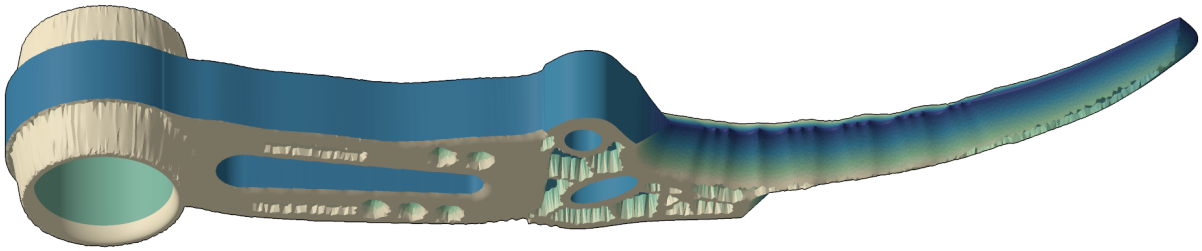
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



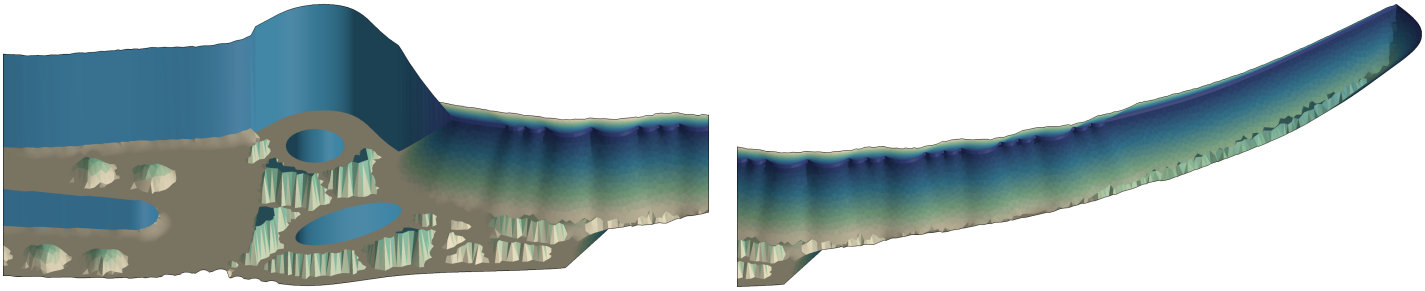
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L3 10 iterations, feature pins 30°

$F = 75.327832$ (with B-rep kept: 80.327832) max displacement 0.194 mm, RMS 0.0123 mm tip SF 0.6877 3D mass 83.1758 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



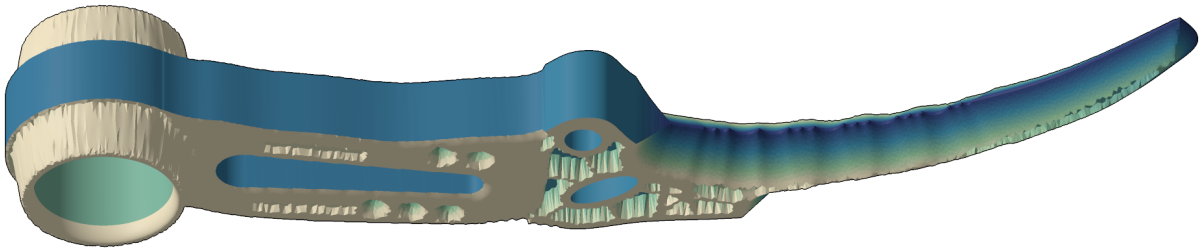
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



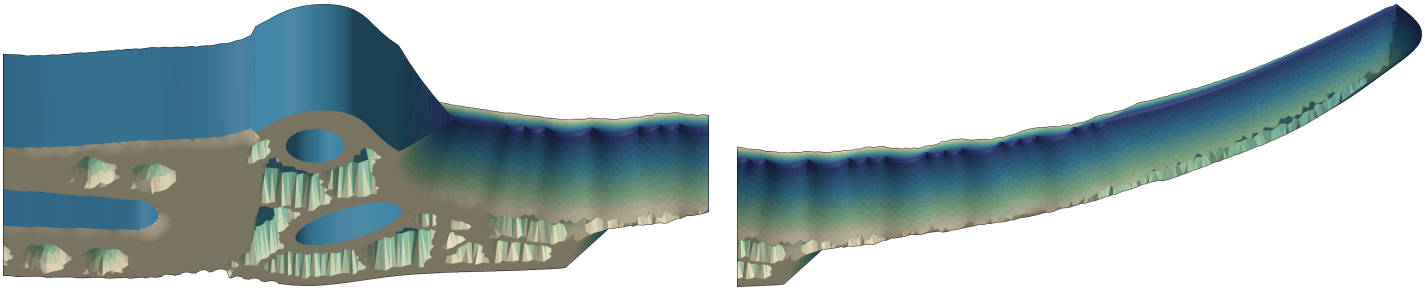
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L4 20 iterations, feature pins 30°

$F = 75.343635$ (with B-rep kept: 80.343635) max displacement 0.214 mm, RMS 0.0156 mm tip SF 0.6905 3D mass 83.1472 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



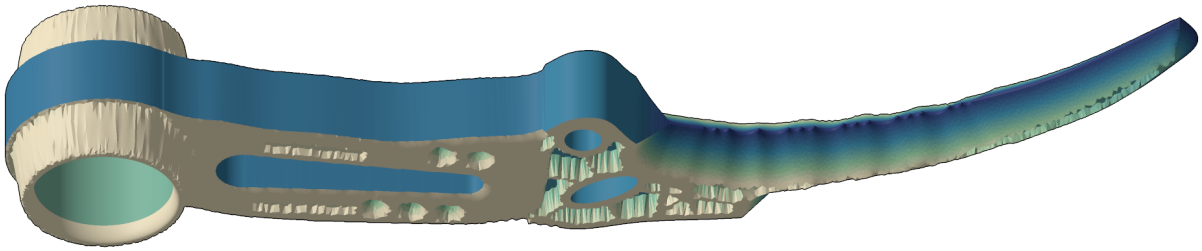
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



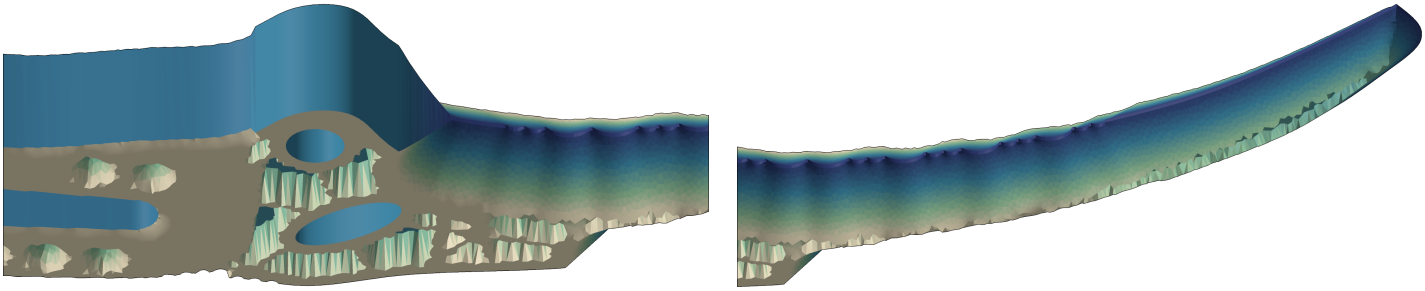
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L5 60 iterations, feature pins 30°

$F = 75.185235$ (with B-rep kept: 80.185235) max displacement 0.252 mm, RMS 0.0222 mm tip SF 0.6333 3D mass 83.0962 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



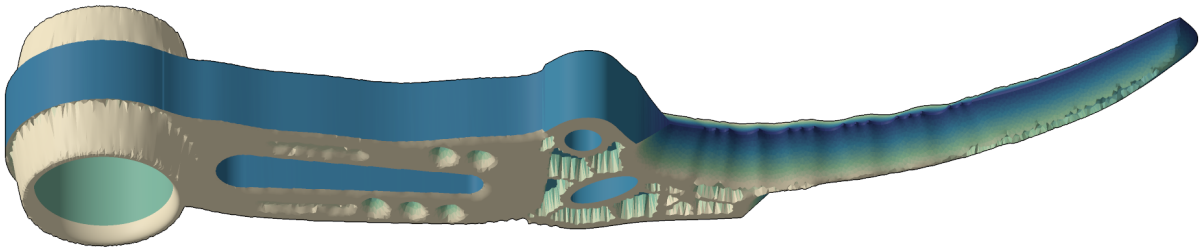
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



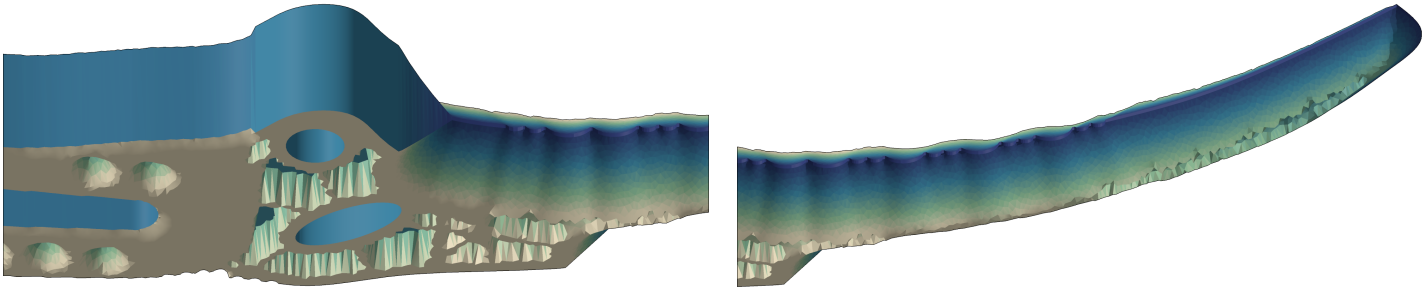
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L6 20 iterations, feature pins 60°

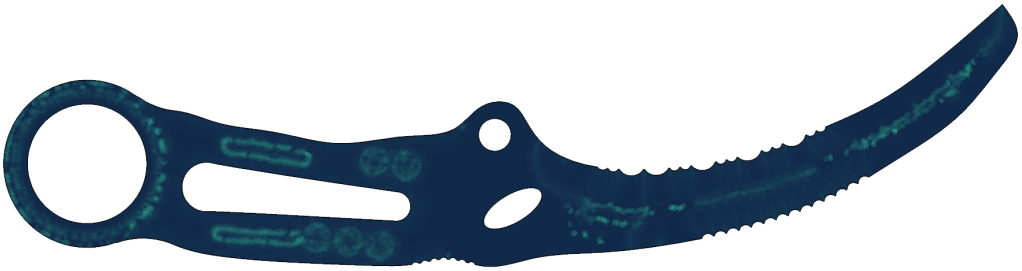
$F = 75.301958$ (with B-rep kept: 80.301958) max displacement 0.423 mm, RMS 0.0361 mm tip SF 0.6842 3D mass 83.2352 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



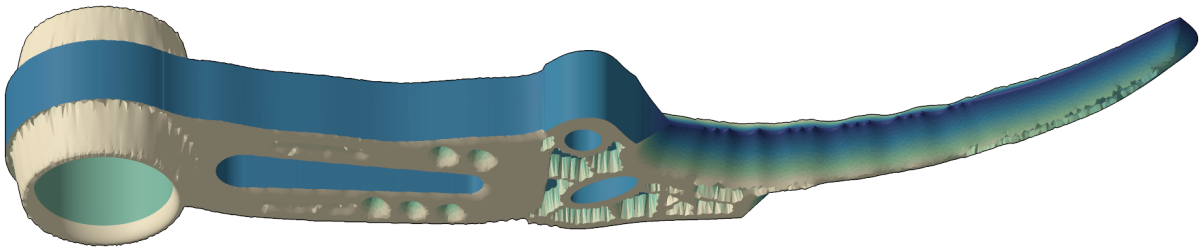
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



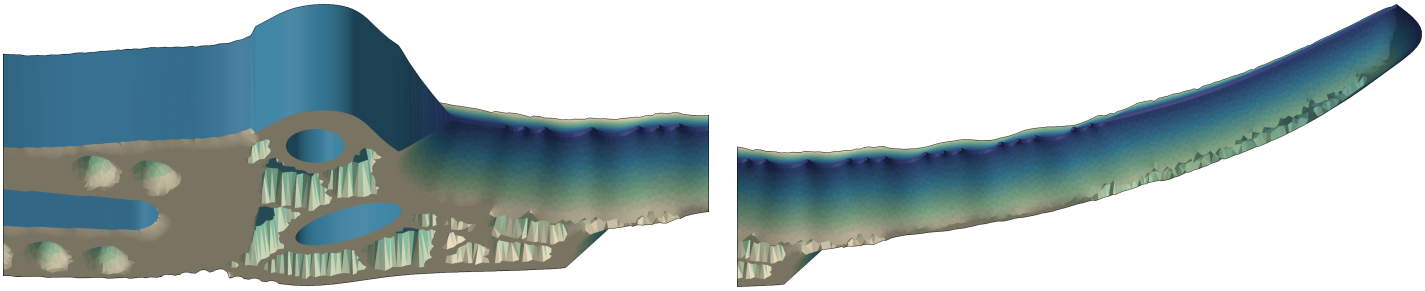
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L7 60 iterations, feature pins 60°

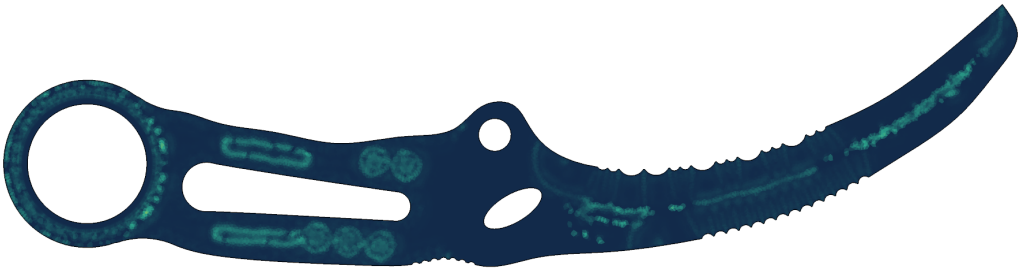
$F = 75.290069$ (with B-rep kept: 80.290069) max displacement 0.495 mm, RMS 0.0502 mm tip SF 0.6750 3D mass 83.1745 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



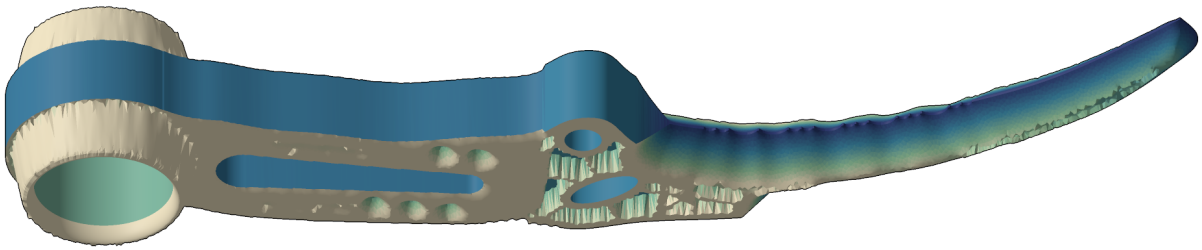
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



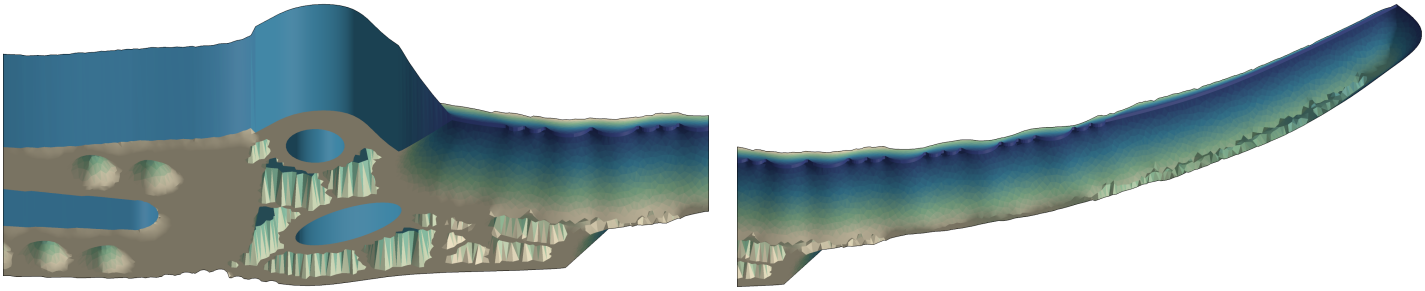
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L8 120 iterations, feature pins 60°

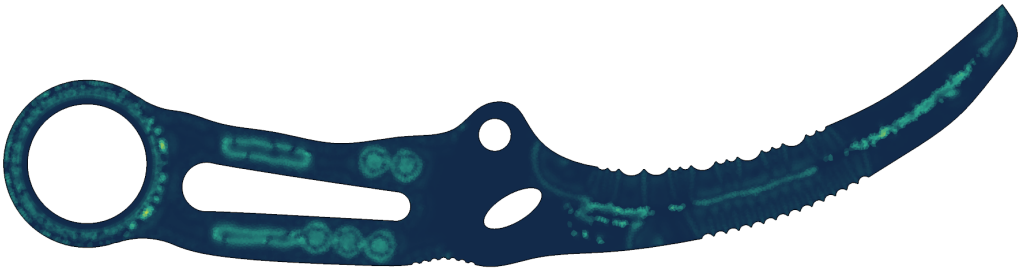
$F = 75.097738$ (with B-rep kept: 80.097738) max displacement 0.613 mm, RMS 0.0636 mm tip SF 0.5997 3D mass 83.0448 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



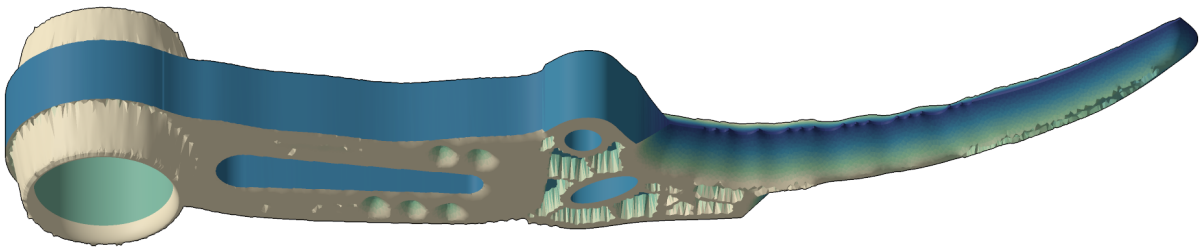
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



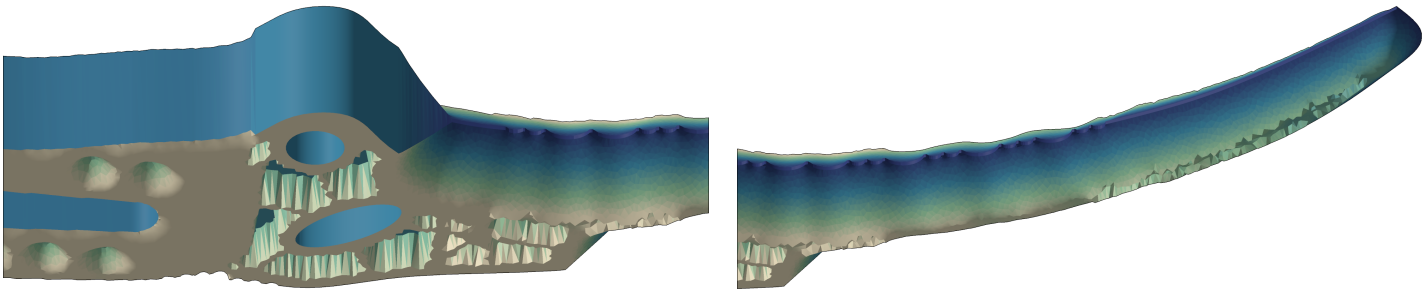
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L9 200 iterations, feature pins 60°

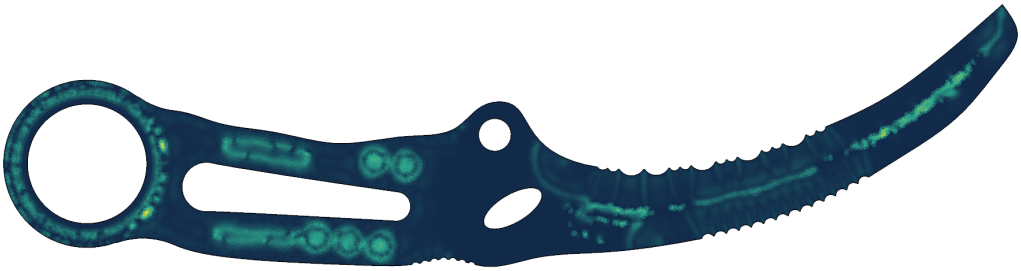
$F = 75.062571$ (with B-rep kept: 80.062571) max displacement 0.703 mm, RMS 0.0766 mm tip SF 0.5747 3D mass 82.8909 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.